

LIPID EXCIPIENTS: CARVING OUT A NICHE IN PHARMACEUTICAL DRUG DELIVERY

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Introduction:

In order to harvest the maximum therapeutic activity of the drug molecules, appropriate dosage forms are designed using various additives called 'Excipients'. It is due to the use of excipients that the drug can be delivered to the patients in a presentable form that meets the targeted dose requirements. Excipients represent diverse classes with versatile structures and are of natural or semi-synthetic origin, small molecules and macromolecules and represent all states of matter, that is, gases, liquids, semi-solid and solids ^[1].

The development of poorly soluble drugs for oral administration presents multi-faceted challenges for formulation scientists including the need to ensure adequate drug solubilization in the physiological environment of the gastrointestinal tract (GI). This is one of the major hurdles for BCS class II drugs that must be overcome to ensure adequate absorption and systemic circulation. It is a well-known fact that majority of New Chemical Entities in the current drug development pipeline are poorly soluble compounds. This explains the pharmaceutical industry's interest in finding robust technologies and solutions to improve the rate of success of new drug development.

Lipid-based drug delivery systems (LBDDS) are one of the emerging technologies designed to address such challenges. Encapsulating or solubilizing the drug in lipid excipients can lead to increased solubilization and absorption, resulting in enhanced bioavailability. Their advantages include dose reduction, elimination of food effect, reduction in first-pass metabolism by facilitating lymphatic route transport and improvement in physical and chemical stability of the drug product. Lipid based drug delivery system can be a simple mixture in oil or a complex self-emulsifying system that spontaneously emulsifies in the presence of aqueous environment. These systems can be formulated as simple oral

liquid solutions or filled into soft or hard gelatine capsules. Recent advances in these formulation technologies have led to the successful commercialization of lipid- based formulations ^[3].

Oleochemistry & Physicochemical properties of Lipid Excipients:

Lipids may be obtained from animal or vegetable sources. Vegetable oils are obtained either from seeds, kernels, or fruits. Each of these species has its unique composition and repartition of fatty structures in terms of hydrocarbon chain length and the number of unsaturated bonds in the chain. These structural variations impact the physical properties of the glycerides. Generally, the term "oil" is used to describe glycerides that are liquid at or above room temperature whereas "fat" is applied to solid or semi-solid glycerides. Fatty acids from vegetable origin are composed of an even number of carbons ranging from 8 to 24. The more commonly recurring fatty acids possess 12, 14, 16, or 18 carbon atoms. Their nomenclature, composition, and melting temperatures are provided in Table 1.

Table 1: Nomenclature and characteristics of some fatty acids.

Common name	Fatty acid chain length and unsaturation ^a	Molecular formula	Melting temp (°C)
Caprylic acid	8:0	CH ₃ -(CH ₂) ₆ -COOH	16.5
Capric acid	10:0	CH ₃ -(CH ₂) ₈ -COOH	31.6
Lauric acid	12:0	CH ₃ -(CH ₂) ₁₀ -COOH	44.8
Myristic acid	14:0	CH ₃ -(CH ₂) ₁₂ -COOH	54.4
Palmitic acid	16:0	CH ₃ -(CH ₂) ₁₄ -COOH	62.9
Stearic acid	18:0	CH ₃ -(CH ₂) ₁₆ -COOH	70.1
Oleic acid	18: 1 (9c)	CH ₃ -(CH ₂) ₇ -CH=CH-(CH ₂) ₇ -COOH	16.0
Ricinoleic acid	18: 1 (9c), OH(12)	CH ₃ -(CH ₂) ₅ -CH(OH)-CH ₂ -CH=CH-(CH ₂) ₇ -COOH	-5.5
Linolenic acid	18: 3 (9c12c15c)	CH ₃ -CH ₂ -CH=CH-CH ₂ -CH=CH-CH ₂ -CH=CH-(CH ₂) ₇ -COOH	-11.0
Eicosenoic acid	20:1 (11c)	CH ₃ -(CH ₂) ₇ -CH=CH-(CH ₂) ₉ -COOH	23.0



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Behenic acid	22:0	$\text{CH}_3-(\text{CH}_2)_{20}-\text{COOH}$	80.0
Erucic acid	22:1	$\text{CH}_3-(\text{CH}_2)_7-\text{CH}=\text{CH}-(\text{CH}_2)_{11}-\text{COOH}$	33.8

a: Number of carbon atoms; number of unsaturated bonds (position and conformation of unsaturation)

c: stands for the cis conformation of the unsaturation bound by opposition to the trans conformation.

Fatty acids are the single common denominator in all lipids. The functionality of lipids is linked to their structural moieties, notably the type of fatty acids and their esters present. Many lipids are amphiphilic in nature, having a lipophilic portion (fatty acid) and a hydrophilic portion. The melting point increases as the fatty acid chain length increases, but it decreases with the increase in the unsaturation of the fatty acid and also increases the susceptibility to oxidation.

A wide range of lipid excipients are available from excipient suppliers. Since these lipids affect the absorption process, it is necessary to know the characteristics of various excipients. The factors that determine the choice of excipients for lipid-based formulations include miscibility; solvent capacity; self-dispersibility and ability to promote self - dispersion of the formulation; digestibility and fate of digested products; regulatory issues – irritancy, toxicity, purity, chemical stability; capsule compatibility; melting point, and cost [3, 4 & 8].

Lipid Formulation Classification System:

Lipid-based delivery systems range from simple oil solutions to complex mixtures of oils, surfactants, co – surfactants and cosolvents. The latter mixtures are typically self-dispersing systems often referred to as self-emulsifying drug delivery systems (SED DS) or self-microemulsifying drug delivery systems (SMED DS). A wide range of triglycerides, partial glycerides, semi-synthetic oily esters, and semi-synthetic non-ionic surfactants esters are available from excipient suppliers.

The main purpose of the LFCS is to enable in vivo studies to be interpreted more readily, and subsequently to facilitate the identification of the most appropriate formulations for specific drugs, i.e. with reference to their physicochemical properties. Table 2 indicates the fundamental differences between Type I, II, III and IV formulations. Whilst the defining properties of each Type are easy to understand, as yet there are few studies which link the LFCS with in vitro or in vivo performance.

Table 2: Lipid Formulation Classification System with indicative excipient composition for each type of formulation

Formulation	Composition		Formulation characteristics in aqueous dispersion
Type I	Oils	100 %	Non-dispersible.
Type II	Oils	40-80%	SED DS without water-soluble components. Opaque emulsion (200 nm-10 µm)
	Low HLB surfactants	20-60%	

Type III	Oils	< 20-80%	SED DS/SMED DS with water-soluble components. Ultrafine dispersion (10-200 nm).
	High HLB surfactants	20-50%	
	Hydrophilic cosolvents	0-50%	
Type IV	Low HLB surfactants	0-20%	Transparent micellar solution/nanosystem (2-10 nm).
	High HLB surfactants	30-80%	
	Hydrophilic cosolvents	0-50%	

Type I formulations contain only lipophilic ingredients (oils) that are non-dispersible in aqueous media. The lipid excipients are digested in vivo forming mixed micelles.

Type II and III formulations contain miscible excipients which form a homogenous dispersion on contact with aqueous media (ie. self-emulsifying system). The particle size and appearance of the dispersion (emulsion or micro/nano emulsion) depends on the properties of the surfactant and the oil phase concentration.

Type IV formulations contain hydrophilic components: water soluble surfactants and hydrophilic co-solvents. They form transparent micellar solutions on contact with aqueous media [2, 3, 6 & 7].

Factors affecting the choice of excipients for lipid-based formulations

- Solvent capacity
- Miscibility
- Morphology at room temperature (i.e. melting point)
- Self-dispersibility and role in promoting self-dispersion of the formulation
- Digestibility, and fate of digested products
- Capsule compatibility
- Purity, chemical stability
- Regulatory issues—irritancy, toxicity, knowledge and experience
- Cost of goods

A Perspective on Safety and Digestibility of Lipid Excipients:

Lipid formulations systems increase absorption from the gastrointestinal tract by accelerating the dissolution process, facilitating the formation of solubilized phases by reduction of particle size to the molecular level, yielding a solid-state solution within the carrier changing drug uptake, efflux and disposition by altering enterocyte-based transport and enhancing drug transport to the systemic circulation via intestinal lymphatic system.

Toxicity is an important issue with regard to the choice of surfactants. Water insoluble surfactants penetrate and fluidize biological membranes and water-soluble surfactants have the potential to solubilize membrane components. All surfactants are potentially irritant or poorly tolerated as a result of these non-specific effects. In general terms cationic surfactants are more toxic than anionic surfactants which in turn are more toxic than non-ionic surfactants. Lipid-based delivery systems usually only include non-ionic surfactants so it is pertinent to compare the

toxicity of non-ionic surfactants. In general bulky surfactants such as polysorbates or polyethoxylated vegetable oils are less toxic than single-chain surfactants, and esters are less toxic than ethers (which are non-digestible). Non-ionic surfactants are generally considered to be acceptable for oral ingestion, and the emergence of several successful marketed products has given the industry confidence in lipid-based products

After oral administration of lipid-based formulations, gastric lipase initiates the digestion of exogenous dietary triglyceride (TG) and formulation TG. Simultaneously, the mechanical mixing (propulsion, grinding and retropulsion) of the stomach facilitates formation of a crude emulsion (comprised of aqueous gastric fluid and lipid digestion products). Later in the small intestine, TG is broken down to diglyceride, monoglyceride and fatty acids by pancreatic lipase. The presence of exogenous lipids in the small intestine stimulates the secretion of endogenous biliary lipids from the gallbladder, including bile salt (BS), PL and cholesterol. Previously formed monoglycerides, fatty acids, and lysophospholipid (products of lipid digestion) are subsequently incorporated into a series of colloidal structures, including micelles and unilamellar and multilamellar vesicles in the presence of bile salts. The solubilization and absorptive capacity of the small intestine for lipid digestion products and drugs (D) is significantly enhanced due to these formed lipid metabolites.

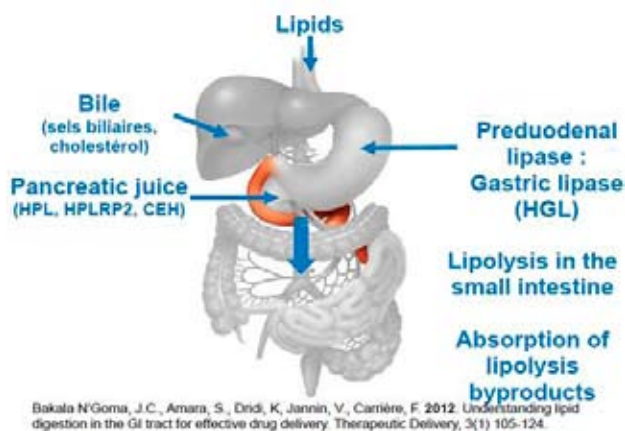


Fig. 1: Understanding the Food Effect on digestion of LBDDS

In particular, it is the lipid component of the food that plays a vital role in the absorption of lipophilic drugs, leading to enhanced oral bioavailability. This can be explained by the ability of a high fat meal to stimulate biliary and pancreatic secretions, to decrease metabolism and efflux activity, to increase intestinal wall permeability, and to a prolongation of gastrointestinal tract (GIT) residence time and transport via lymphatic system ^[3 & 5].

Characterization & Evaluation of the performance of Lipid Excipients:

i. Characterization of lipid systems:

Physical analysis: Analysis of the thermal behaviour of lipids during formulation is of primary importance, since lipid excipients have complex chemical compositions that lead to broad melting ranges. Various thermal properties of lipids including crystallization temperature, melting point, glass transition temperature and determination of solid fat content of the excipient versus temperature can be evaluated using differential scanning

calorimetry (DSC). The organization of the lipid during heating or cooling can be assessed by hot-stage microscopy. Crystallinity of a lipid excipient can be confirmed by X-ray diffraction (XRD).

Chemical analysis: High performance liquid chromatography (HPLC) and gas chromatography (GC) can be used to determine the exact composition of ethers, esters, and fatty acid distribution. Other chemical indices are also available; the molecular weight of fatty acids can be determined from their saponification value; the saturation of hydrocarbon chains can be measured using an iodine-based assay; oxidative changes can be determined by measuring peroxides; free fatty acids can be measured from acid content; and free hydroxyl groups can be determined by measuring hydroxyl group content.

ii. Mutual miscibility of excipients:

Mutual miscibility of excipients is necessary to produce a clear, stable, liquid formulation. The chemical diversity of lipid excipients can lead to immiscibility on long-term storage, so long-term physical stability tests should be carried out routinely during commercial development projects. Waxy excipients often need to be heated to ensure adequate mixing of all components before the excipient is weighed for blending. This can lead to dissolution of saturated fatty acid components at elevated temperature, leading to supersaturation at ambient temperature. Re-crystallization of the waxy components may take weeks or months once excipients have been blended. Temperature cycling tests may help to identify potential problems with supersaturation, but anticipation of potential problems is vital.

iii. Solvent capacity:

Triglycerides are poor solvents for all but highly lipophilic compounds, so most lipid-based formulations contain polar oils, surfactants and/or cosolvents to improve the solvent capacity of the anhydrous formulation. The formulator must balance the advantage of including cosolvents with the risk of inducing drug precipitation on dispersion.

iv. Capsule compatibility:

Low molecular weight polar molecules present in capsule formulations are able to penetrate and plasticize gelatin capsule shells, which restricts the concentration of propylene glycol and related cosolvents that can be used in capsule fills. Surfactants can also destabilise capsule shells but there are differences between soft and hard gelatin capsules. Capsule compatibility problems are well-known to capsule manufacturers and should be a part of stability study.

v. In vitro studies:

A preliminary guideline for formulation development and assessment of drug release can be obtained from in vitro studies. Apart from assessing batch-to-batch consistency, these studies also depict the in vivo evaluation of the lipid-based formulation.

a. Self-dispersion and sizing of dispersions:

Generally well-formulated SEDDS or SMEDDS are dispersed within seconds under conditions of gentle stirring, and visual observation to distinguish good and poor formulations may be adequate. Poor self-emulsifying formulations are technically challenging to size because typically they are coarse, polydisperse, and sensitive to dilution and agitation. Usually, a poor formulation can be distinguished from an adequate formulation by determining polydispersity using a Fraunhofer

diffraction sizer. Optimization of SEDDS and SMEDDS is more likely to be a job for a photon correlation sizer, but if a Fraunhofer instrument is available it is advisable to use this instrument to check that there are no particles larger than 1 μm .

b. Dissolution in biorelevant media:

The FDA requires conventional USP dissolution testing as a quality control tool for the formulations, but this dissolution does not correlate to the *in vivo* behaviour of lipid-based formulations. In the GIT lipids are subjected to digestion processes in the presence of gastric and pancreatic lipases. These lipases also affect the emulsification and dispersion properties of the lipid excipients, leading to altered solubilization capacity *in vivo*. To assess such effects and predict *in vivo* behaviour, dissolution testing in biorelevant media can be helpful.

c. In vitro dispersion test:

The *in vitro* dispersion test assesses the likelihood of drug precipitation as a LBF disperses within gastric fluid before entering the small intestine. Such tests are routinely performed prior to digestion tests in the development of LBFs as they provide the opportunity to rapidly screen-out formulations that precipitate on simple dilution.

Dispersion testing can be carried out using a standard dissolution apparatus but, assuming that the drug is initially in solution in the anhydrous formulation, the emphasis should not be on dissolution but rather on detecting unwanted precipitation of the drug.

d. In vitro digestion test:

Lipid-based excipients are known to enhance the oral absorption of drugs by affecting various physiological processes. These include stimulating bile flow and pancreatic juice secretion, prolonging gastric emptying, increasing the membrane fluidity, opening of tight junctions, promoting lymphatic transport of drugs, thus avoiding first pass metabolism and inhibiting efflux transporters.

Physiological digestive enzymes digest fats (triglycerides) by the hydrolysis of ester bonds that release free fatty acids and monoglycerides. Lipid excipients comprise fatty acid esters (glycerides, PEG esters, etc.) and are therefore subject to hydrolysis by the same endogenous enzymes. The degree of hydrolysis of a lipid excipient (lipolysis) can be measured *in vitro* by analysing free fatty acid levels after a period of enzymatic digestion.

The *in vivo* digestion of lipid excipients can positively contribute to the performance of LBFs by promoting drug solubility in mixed micelles and by presenting drug in a solubilized state at the GI membrane.

Digestion testing is of even greater significance because it offers the opportunity to predict the fate of the formulation and drug in the intestinal lumen prior to absorption.

e. In vivo Studies:

The impact of excipients on the bioavailability and pharmacokinetic profile of drugs can be estimated by designing

appropriate *in vivo* studies. A detailed study of intestinal lymphatic absorption is required since lipid-based formulations enhance bioavailability by improving the intestinal uptake of drug [3, 5 & 7].

Regulatory status of lipid excipients:

The U.S. FDA deems drug products therapeutic equivalents if they are pharmaceutical equivalents and can be expected to have the same clinical effect and safety profile when administered to patients under the conditions specified in the labelling. The unique characteristics for lipid excipients/surfactants and lipid-based dosage forms, additional scrutiny of formulation and potential biopharmaceutical interactions are necessary for adequate evaluation of pharmaceutical equivalence and bioequivalence. A difference in the makeup of lipid formulations may result in a significant impact on the outcome of bioequivalence. The recent FDA's critical path initiative offers an excellent opportunity for enhancing regulatory sciences and fostering development of novel dosage forms such as lipid-based delivery systems [2, 3, 5, 6, 7].

Conclusion:

Recent advances in Lipid-based drug delivery systems (LBDDS) have led to the successful commercialization of lipid-based formulations. LBDDS could be aptly utilized for differentiated product development and supgeneric perspective for product lifecycle management. Moreover, development of regulatory guidelines for lipid-based formulations will further enrich the advancement of this technology. Further research has to be carried out in this field regarding the design of a proper *in vivo* model to correlate the data obtained *in vitro* studies to the actual *in vivo* experience.

References:

1. Panayiotis Constantinides, Subhashis Chakraborty and Dali Shukla; Considerations and recommendations on traditional and non-traditional uses of excipients in oral drug products; AAPS Open (2016) 2:3; DOI 10.1186/s41120-016-0004-3.
2. Mei-Ling Chen; Lipid excipients and delivery systems for pharmaceutical development: A regulatory perspective; Advanced Drug Delivery Reviews 60, (2008); 768–777.
3. Sandeep Kalepu, Mohanvarma Manthina, Veerabhadraswamy Padavala; Oral lipid-based drug delivery systems – An overview; Acta Pharmaceutica Sinica B; 3(6) (2013); 361 – 372.
4. Vincent Jannin, Jean-David Rodier & Jasmine Musakhanian; Polyoxylglycerides and glycerides: Effects of manufacturing parameters on API stability, excipient functionality and processing; International Journal of Pharmaceutics; 466 (2014); 109–121.
5. Colin W. Pouton, Christopher J.H. Porter; Formulation of lipid-based delivery system for oral administration: Materials, methods and strategies; Advanced Drug Delivery Reviews; 60 (2008); 625 – 637.
6. Shweta Gupta, Rajesh Kesarla & Abdelwahab Omri; Formulation Strategies to Improve the Bioavailability of Poorly Absorbed Drugs with Special Emphasis on Self-Emulsifying Systems; (2013); <http://dx.doi.org/10.1155/2013/848043>
7. Hina Shrestha, Rajni Bala & Sandeep Arora; Lipid-Based Drug Delivery Systems; Journal of Pharmaceutics; (2014); <http://dx.doi.org/10.1155/2014/801820>.